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# Auction Platform Development Task

**Overview:**

You are tasked with developing core components of an online auction platform. The platform will allow users to list products (both new and old) for auction, and buyers will be able to place bids to acquire these products at reasonable prices. The primary goal of the platform is to ensure a smooth bidding process where users can engage in competitive auctions while preventing unreasonable pricing.

You will be responsible for building the following features:

- Auction listing and bidding system.
- Time-based auction management.
- User authentication and role-based access control (buyer vs seller).
- Logic to handle auction winners.
- Feedback system for users after the auction ends.

Your code should be efficient, well-structured, and handle various edge cases such as invalid inputs, race conditions in bidding, and more. Test your implementations with various scenarios to ensure correctness and robustness.

## Coding Tasks:

### 1. Create an Auction Listing

- **Objective:** Build an API that allows sellers to create a product listing for auction.
- **Details:** Users will provide details such as the product name, description, starting price, auction start time, and end time.
- **Test:** Validate fields, ensure times are correct, and ensure a new auction is successfully created.

### 2. Place a Bid

- **Objective:** Build an API that allows users to place bids on auctioned items.
- **Details:** Buyers can place bids, but each bid must be higher than the current highest bid and placed within the auction's active time window.
- **Test:** Ensure bids follow the auction rules and handle edge cases where the bid is too low or the auction has ended.

### 3. Auction Timer Management

- **Objective:** Create a system that manages auction states based on the current server time.
- **Details:** Handle the automatic opening and closing of auctions based on the start and end time.
- **Test:** Ensure that auctions only accept bids during their active periods and are correctly marked as closed after the end time.

### 4. User Authentication & Roles

- **Objective:** Implement a basic user authentication system and define roles (buyer/seller).
- **Details:** Ensure only sellers can create auctions, and only buyers can place bids.
- **Test:** Validate login functionality, role assignment, and restrictions.

### 5. Determine Auction Winner

- **Objective:** Automatically determine the highest bidder once the auction ends and mark them as the winner.
- **Details:** Implement logic to select the highest bidder at the auction's close.
- **Test:** Ensure the highest bid is correctly marked as the winner after the auction ends.

### 6. Product Price Validation

- **Objective:** Prevent unreasonable pricing by checking if a product's final bid is below a certain threshold.
- **Details:** Implement a review system that flags auctions where the final price is too low.
- **Test:** Ensure that bids below 50% of the original price are flagged for review.

### 7. User Feedback System

- **Objective:** Allow users to leave feedback after completing an auction.
- **Details:** Both buyers and sellers should be able to leave reviews and ratings for each other after a transaction.
- **Test:** Validate feedback inputs (text length, rating values) and ensure reviews are saved correctly.

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### Submission Procedures:

1. **Code Submission:**
   - Submit your project code via GitHub, Bitbucket, or any version control system of your choice.
   - Ensure your repository is public or provide access to the reviewers.
   - Submit a postman collection with sample requests
   - Include a `README.md` file explaining:
     - How to run your code (dependencies, environment setup).
     - API endpoints and sample requests/responses.
     - Any special instructions or assumptions made.
2. **Code Quality:**
   - Follow standard best practices for clean, maintainable code.
   - Include comments where necessary to explain complex logic.
   - Use consistent code formatting and naming conventions.
3. **Deadline:**
   - Submit the project by the agreed deadline.

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